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DeepNeoAG: Neoantigen epitope prediction from melanoma antigens using a synergistic deep learning model combining protein language models and multi-window scanning convolutional neural networks

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ABSTRACT

Neoantigens, derived from tumor-specific mutations, play a crucial role in eliciting anti-tumor immune responses and have emerged as promising targets for personalized cancer immunotherapy. Accurately identifying neoantigens from a vast pool of potential candidates is crucial for developing effective therapeutic strategies. This study presents a novel deep learning model that leverages the power of protein language models (PLMs) and multi-window scanning convolutional neural networks (CNNs) to predict neoantigens from normal tumor antigens with high accuracy. In this study, we present DeepNeoAG, a novel framework combines the global sequence-level information captured by a pre-trained PLM with the local sequence-based information features extracted by a multi-window scanning CNN, enabling a comprehensive representation of the protein's mutational landscape. We demonstrate the superior performance of DeepNeoAG compared to existing methods and highlight its potential to accelerate the development of personalized cancer immunotherapies.

1. Introduction

Traditional tumor antigens, known as tumor-associated antigens (TAAs) [1], are expressed by both tumor and normal cells. Due to the immune system's central tolerance mechanism, T cell receptors (TCRs) with high affinity for self-antigens, which could potentially cause autoimmune responses, are eliminated. This process leaves behind TCRs with low affinity, which are often ineffective at triggering a strong immune response. When TAAs are targeted for immunotherapy, there is a risk of attacking normal cells, leading to potential autoimmune disorders.

Neoantigens, also referred to as tumor-specific antigens (TSAs) [2], are produced by gene mutations in tumor cells and are not expressed in normal cells. Immune cells can recognize these neoantigens as "non-self," activating T cells and eliciting an anticancer immune response. Since neoantigens are less impacted by immune tolerance, they have a higher likelihood of activating specific TCRs that target tumors, making them particularly attractive targets for immunotherapy.

Neoantigen vaccines have demonstrated safety and efficacy in several clinical trials [3–5], showing great potential in treating various types of cancer. These vaccines work by administering a selected group of neoantigens to the patient. As shown in Fig. 1, neoantigens are processed into small epitope fragments and presented on the surface of dendritic cells via major histocompatibility complex II (MHC II) molecules [6] molecules. This leads to the activation of TCRs and the initiation of a cytotoxic immune response. Due to their high specificity and low side effects, neoantigen vaccines have gained significant attention in recent years as a promising avenue for cancer treatment.

Accurate neoantigen prediction is crucial for the success of cancer immunotherapy, particularly in personalized treatments that harness the immune system to target tumor cells [7,8]. Neoantigens, arising from somatic mutations, are ideal candidates for such therapeutic approaches. Traditional methods for neoantigen prediction focus on evaluating the binding affinity between peptides and MHC molecules. Tools such as NetMHCpan [9] for class I MHC and NetMHCIIpan [9] for class II MHC have significantly improved the accuracy of these predictions by training machine learning models on peptide sequences and binding affinity data. However, predicting MHC class II binding affinity remains more challenging due to variability in peptide lengths and the complex structure of class II molecules. Although advancements have been made by tools like NetMHCIIpan [9], MixMHC2pred [10] and DeepSeqPanII [11], limitations persist, especially in predicting

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neoantigens for rare or poorly characterized MHC alleles.

To address these challenges, our method, DeepNeoAG, employs deep learning techniques, including multi-window scanning and protein language models (PLMs), to extract features directly from peptide sequences, bypassing the need for explicit MHC allele input. By utilizing PLMs' ability to learn complex sequence features, DeepNeoAG can predict neoantigens even when MHC data is incomplete or unavailable. This flexibility enables DeepNeoAG to complement existing neoantigen prediction tools and fill gaps, especially for rare MHC alleles or novel tumor types. Although MHC binding affinity remains a vital component, DeepNeoAG's approach offers a broader, more accessible path for neoantigen identification in cancer immunotherapy and can be integrated with traditional affinity models in future research.

2. Materials and methods

2.1. Data collection

We compiled two relevant datasets from the CEDAR database [12], which contains experimental data on T cell epitopes related to melanoma. As shown in Table 1. The first dataset consisted of 671 confirmed neoantigen epitopes, which were labeled as positive samples. The second dataset contained 32,206 epitopes derived from viral antigens, and tumor-associated antigens (TAAs), which were labeled as negative samples.

Next, we applied CD-Hit [13] to remove peptides with >30 % sequence similarity to reduce redundancy and focus the model training on a more diverse set of epitopes. This 30 % threshold is commonly used in bioinformatics as it helps strike a balance between maintaining sequence diversity and retaining relevant structural similarities. By filtering out highly redundant or near-duplicate sequences, we aimed to improve the model's ability to generalize and capture distinguishing features between neoantigens and non-neoantigen epitopes, rather than relying on trivial similarities.

Table 1
The statistics of the survey dataset.

Dataset	Epitopes	Similarity <30 %	Length <35
Neoantigen	671	303	302
Others	32,206	6717	6710
Total	32,877	7020	7012

Dataset	Training data (80 %)	Independent data (20 %)
Neoantigen	241	61
Others	5367	1343
Total	5608	1404

Independent dataset	MHC-I presented peptides	MHC-II presented peptides
Neoantigen	18	43
Others	1005	338
Total	1023	381

We also set a maximum length threshold of 35 amino acids for the peptide sequences to ensure that the model could efficiently learn the relevant patterns while avoiding overly long sequences that may introduce noise or unnecessary complexity. Peptides longer than 35 amino acids may complicate the model's learning process and reduce its ability to focus on essential features for accurate neoantigen prediction. This length threshold was determined based on prior studies in the field, as well as the characteristics of the CEDAR dataset used in the analysis.

The final dataset consisted of 302 neoantigen epitopes and 6710 non-neoantigen epitopes (viral and TAA). We then split this curated dataset into training and independent testing sets with 80:20 ratio. The training set contained 241 neoantigen epitopes, while the independent test set

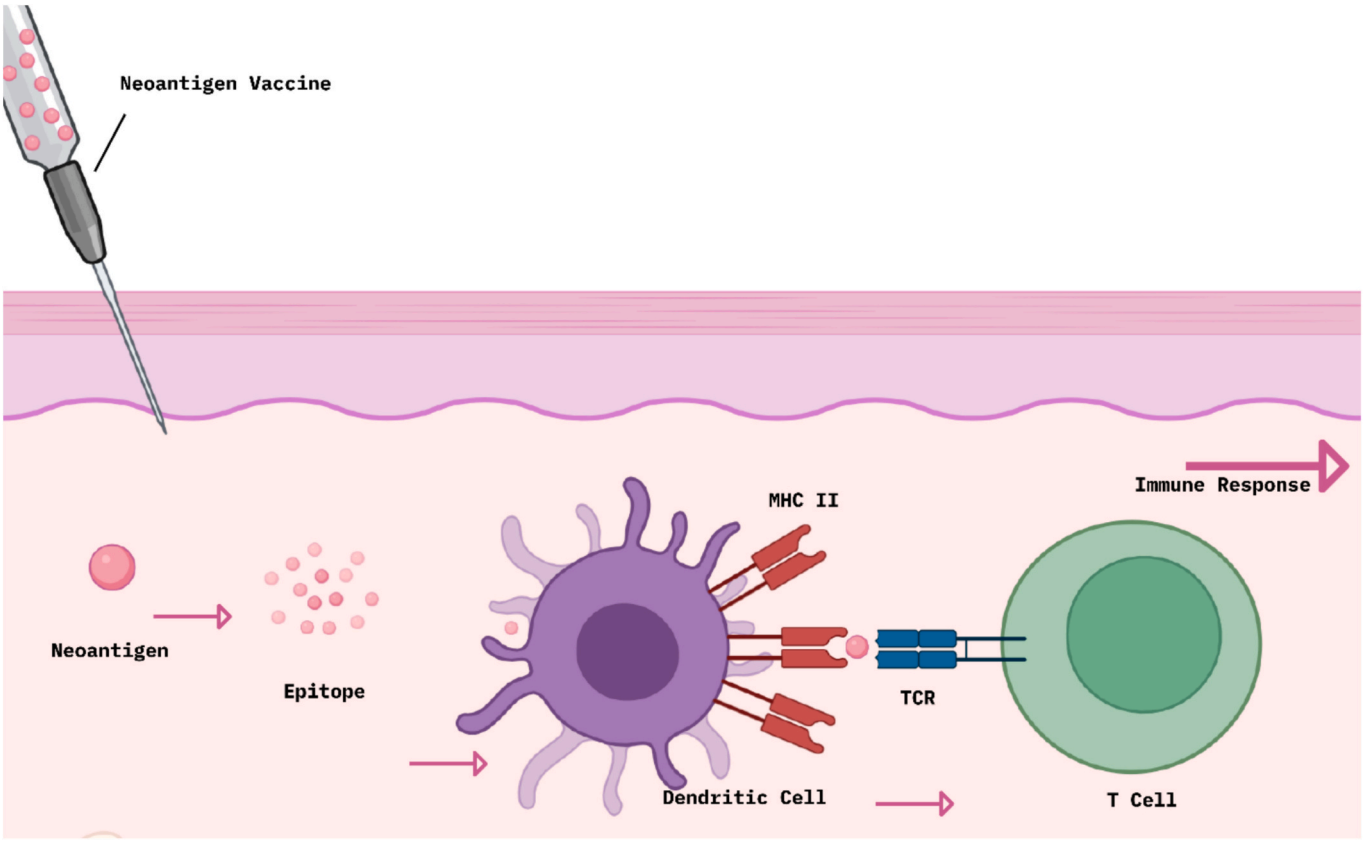


Fig. 1. Cellular mechanisms underlying the action of neoantigen vaccines.

included 61 neoantigen epitopes and 1343 non-neoantigen epitopes.

2.2. Feature extraction

To extract latent information from protein amino acid sequences, we used pre-trained language models based on protein databases. Unlike traditional feature extraction methods, such as multiple sequence alignment [14] and one-hot encoding, the pre-trained language models offer a more sophisticated approach. In traditional methods, similarity calculations or binary encoding are based solely on the sequence. However, pre-trained language models benefit from training on extensive protein databases that encompass a wide range of species and functional proteins.

In these language models, protein amino acid sequences are processed through the encoder of a Transformer [15] model, allowing it to consider contextual information and weigh amino acids in different positions. This enables the model to autonomously learn the relationships and importance of amino acids, leading to a more nuanced understanding of the sequence. In this study, we focused solely on sequence-based features, as peptide sequences are the primary determinants of antigen-TCR interactions and immunogenicity. By leveraging the comprehensive knowledge encoded in pre-trained PLMs, we hypothesized that the model could implicitly capture relevant structural patterns related to neoantigen identification, without requiring explicit structural feature engineering.

Our research involved the use of three pre-trained language models: ProtTrans [16], ESM [17], and TAPE [18], focusing on ProtTrans primarily.

2.2.1. ProtTrans embedding

ProtTrans [16] is a computational tool that leverages advanced language models (such as Transformer-XL [19], XLNet [20], BERT [21], Albert [22], Electra [23], and T5 [24]) to analyze and interpret protein sequences. By analyzing single sequence embeddings, ProtTrans is capable of predicting protein secondary structures, subcellular locations, and other properties. And the tokenization process in ProtTrans involves converting protein sequences into tokens that can be processed by the model, allowing it to capture intricate patterns and relationships within the sequence.

In this study, we used ProtT5, which is a T5 [24] model that has been pre-trained on UniRef50 [25] and generates 1024-dimensional embeddings. The ProtT5 architecture utilizes the strengths of the T5 architecture to handle the complexity of protein sequences, enabling accurate predictions and insights without the dependence on multiple sequence alignments or evolutionary data. This model excels in providing detailed contextualized embeddings, making it a powerful tool for various applications in computational biology and bioinformatics.

2.2.2. ESM embedding

ESM (Evolutionary Scale Modeling) [17] is a cutting-edge computational method for analyzing and interpreting protein sequences that uses advanced linguistic models, specifically Transformer [26] architectures. Using massive protein databases, these models are capable of predicting a wide range of protein properties, such as secondary structures, residue-residue contacts, and mutational effects. In ESM models, such as ESM-2 and ESM-1b, self-supervised learning techniques are used to extract meaningful features from protein sequences without the need for multiple alignments or evolutionary profiles.

In this study, we used the 1280-dimensional ESM-2 and ESM-1b models. In order to generate rich, high-dimensional embeddings that encapsulate the intricate patterns and relationships inherent in protein sequences, these models were pre-trained on comprehensive datasets such as UniRef50 [25]. The ESM models represent a significant advancement in protein sequence analysis, offering superior performance and efficiency compared to traditional methods.

2.2.3. Tape embedding

TAPE (Tasks Assessing Protein Embeddings) [18] is a comprehensive benchmark suite that measures the performance of machine learning models on protein sequence analysis tasks. As a result of the high cost incurred in acquiring supervised protein labels, TAPE includes five biologically relevant semi-supervised learning tasks: secondary structure prediction, contact prediction, remote homology detection, fluorescence landscape prediction, and stability landscape prediction. The tasks are selected to ensure that they evaluate biologically meaningful generalizations that can be applied to real-life scenarios.

We have implemented the BERT-base model from the TAPE family, which is based on Google's BERT [21] model architecture. Based on the UniRef50 [25] protein database, this model provided 768-dimensional embeddings that provide high resolution representations of the original amino acid sequences.

2.3. Multiple window scanning deep learning networks architecture

The multi-window scanning method was first introduced for sentence classification tasks using CNNs [27]. Deepfam [28] modified this approach in 2018 to analyze protein sequences. By 2022, the MCNN-ETC [29] model had successfully incorporated multi-window scanning with attributes derived from multiple sequence alignments. In our study of DeepNeoAG, we have further refined and extended the use of multi-window convolution filters by integrating them with protein language models (PLMs) to improve performance compared to traditional CNN approaches.

Due to the differences in sequence and length characteristics between peptides presented by MHC-I (8–11 amino acids) versus MHC-II (12–25 amino acids). We have employed the multi-window scanning technique, which is a key component of the MCNN model that was adapted for the DeepNeoAG framework to obtain functional information about proteins from sequences. By incorporating embeddings from pre-trained protein language models, including Lx1024(ProtTrans), Lx1280 (ESM-1b,ESM-2), Lx768(Tape), in which L represents the length of input peptides. By combining this multi-window scanning technique with the pre-trained protein language model embeddings, this technique provides a more detailed biological information than what is possible with sequences alone and has been shown to result in better results compared to traditional CNN approaches in recent research.

Next, we put the obtained embedding matrices from pre-trained protein language models through a series of convolutional filters (or windows), which scan these embedding features across various window sizes W. By encoding the contextual environment of each amino acid, this process identifies predictive local motifs that are indicative of neoantigen peptides, which results in an output sequence of length L-W + 1 from each filter as shown in Fig. 2. Stacking filters of varying window sizes allows the model to adeptly merge sequence patterns on both a local and a global level. With this approach, information from both the immediate and broader context of a sequence can be seamlessly integrated, enabling a more comprehensive analysis of the sequence data.

In the final step, instead of forcing all peptide sequences to a fixed length, the MCNN model applies 1-max pooling for each convolutional window and each filter as shown in Fig. 3. This allows the model to extract the most salient features from each window, regardless of the overall sequence length. From these pooling operations, a flattened output is obtained from each convolution window and then concatenated into a vector representing the cross-scale patterns in the sequence. As a result of this combined vector, a fully connected layer is used to perform the final classification task of identifying neoantigen epitopes from normal antigen epitopes without being constrained by the differences in length between the MHC-I and MHC-II presented peptides.

2.4. Performance evaluation

The model's performance was evaluated using several established

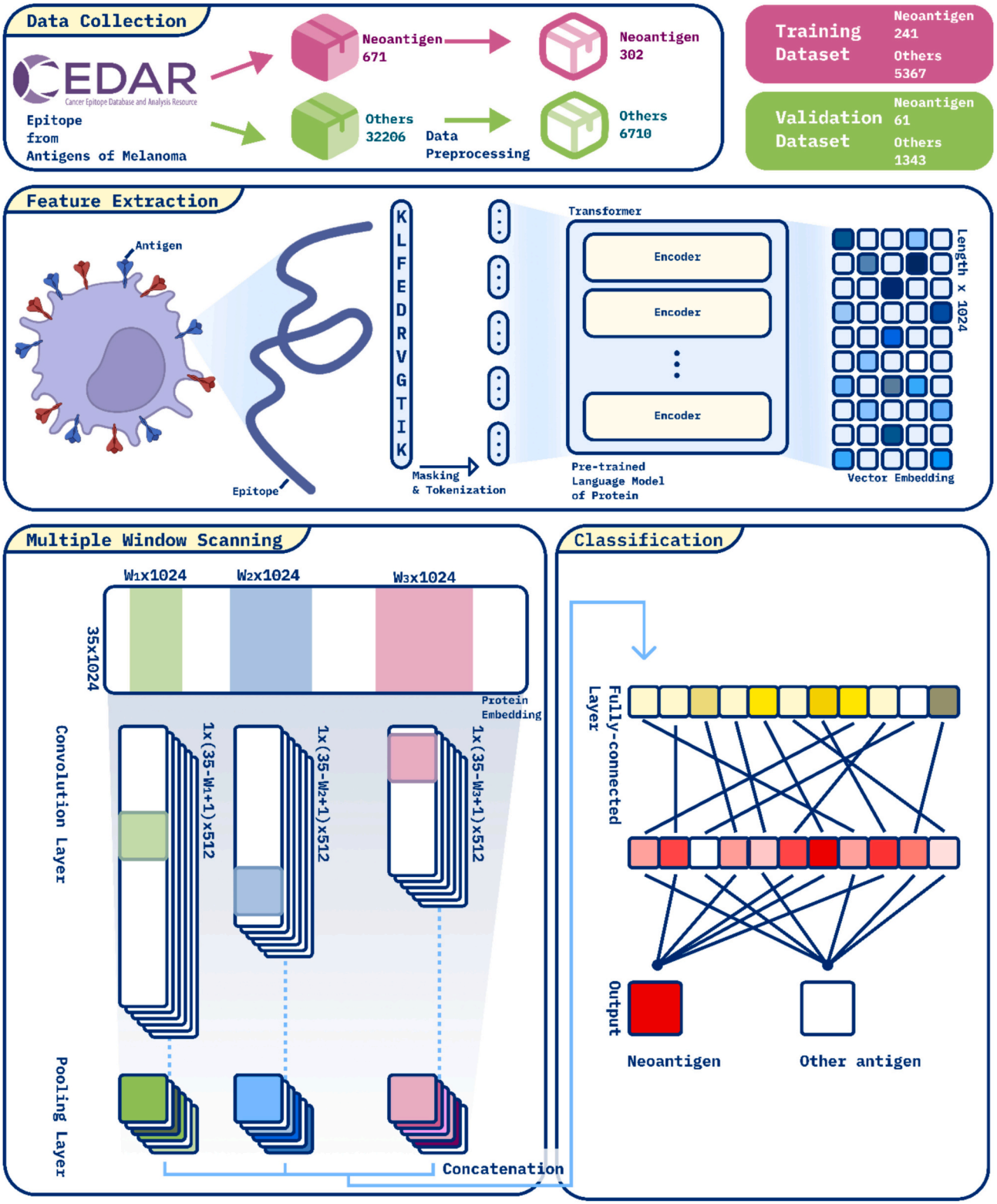


Fig. 2. The workflow for neoantigen epitope prediction model.

metrics, focusing on the five categories of predictions. These evaluation metrics included sensitivity, specificity [30], accuracy, the Matthews correlation coefficient (MCC), and the area under the receiver operating characteristic curve (AUC) [31]. These metrics were derived using formulas 1–4, described below:

$$\text{Sensitivity} = \left(\frac{TP}{TP + FN}\right) \times 100 \tag{1}$$

$$\text{Specificity} = \left(\frac{TN}{TN + FP}\right) \times 100 \tag{2}$$

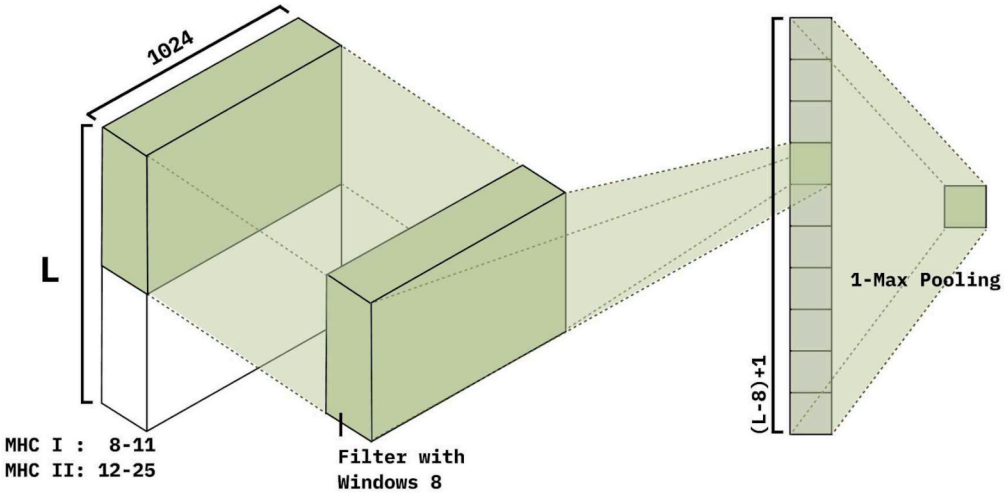


Fig. 3. The flowchart of 1-max pooling with different sequence length.

$$\text{Accuracy} = \left(\frac{TP + TN}{TP + TN + FP + FN}\right) \times 100 \tag{3}$$

$$\text{MCC} = \left(\frac{(TP \times TN) - (FP \times FN)}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}\right) \times 100 \tag{4}$$

FP stands for false positive, FN stands for false negative, TP stands for true positive, and TN stands for true negative in this context.

Additionally, the area under the receiver operating characteristic curve (AUC) values range from 0 to 1, with higher values indicating superior model classification precision, and lower values indicating inferior performance.

3. Results

The performance of identifying neoantigen epitopes from normal antigen epitopes was evaluated using sensitivity, specificity, accuracy, Matthews correlation coefficient (MCC), and area under the ROC curve (AUC). The multi-window CNN model was trained using 5-fold cross-validation and tested on a dataset compiled from CEDAR.

3.1. Performance comparison using different single window sizes

At first, CNN models were developed by incorporating features from ProtTrans, following a single-window approach with window spans ranging from 8 to 34 residues as shown in Table 2. A smaller window size of 8 achieved an MCC of 0.3298 and an AUC of 0.8919, whereas larger windows, such as those spanning 34 residues, yielded a higher MCC (0.5298) and AUC (0.9278). This indicates that larger window sizes are more effective at capturing broader patterns within the data but may reduce the emphasis on detailed local interactions.

3.2. Performance comparison using different window combinations

Following this, multiple window sizes were combined using a hierarchical approach, starting with 34 residues and incrementally adding smaller windows, as shown in Table 3. As a result of adding 32, 24, and 16 windows in addition to the baseline of 34, the MCC was improved to 0.5322 and the AUC was improved to 0.9339. By utilizing a multi-window architecture, we are able to integrate local and global patterns in an advantageous manner.

3.3. Performance comparison using different filter numbers

In addition, the number of filters applied to each window size also affects the performance of the model. As the number of filters is adjusted between 128 and 2048, Table 4 illustrates how the model's performance changes. The AUC for the minimal configuration of 128 filters was 0.9298, while the AUC for the excessive configuration of 2048 filters was 0.9299, possibly due to overfitting. In terms of AUC, 512 filters per window provided the best results with 0.9339.

3.4. Performance comparison using different features

To compare the performance of different pre-trained language models on the multi-window CNN model, we adapted several pre-trained language methods to obtain different feature embeddings in addition to using ProtTrans embeddings. These methods included ESM-2 and ESM-1b transformer protein language models from the Meta Fundamental AI Research Protein Team (FAIR) and the Tasks Assessing Protein Embeddings (TAPE). As shown in Table 5, ProtTrans features outperformed the others, achieving the highest AUC of 0.9339 in cross-validation and an AUC of 0.9323 in independent testing. This indicates that ProtTrans optimally fits our model, making it the most suitable choice for our multi-window CNN model. Fig. 4a, shows the ROC curve of multi-window CNN using different input feature representations.

3.5. Performance comparison using different classifiers

As shown in Table 6, the multi-window ProtTrans-CNN model

Table 2
The performance comparison with different single window sizes.

Window	Sensitivity	Specificity	Accuracy	MCC	AUC
8	0.8565	0.8104	0.8124	0.3298	0.8919
16	0.8397	0.8657	0.8648	0.3988	0.9085
24	0.8532	0.9163	0.9137	0.4941	0.9202
32	0.8616	0.9256	0.9229	0.5182	0.9274
34	0.8593	0.9307	0.9276	0.5298	0.9278

Table 3
The performance comparison with different window combinations.

Multiple window combination	Sensitivity	Specificity	Accuracy	MCC	AUC
[34]	0.8593	0.9307	0.9276	0.5298	0.9278
[34, 32]	0.8495	0.9287	0.9254	0.5247	0.9229
[34, 32, 24]	0.8706	0.9185	0.9165	0.5012	0.9299
[34, 32, 24, 16]	0.8578	0.9322	0.9192	0.5322	0.9339
[34, 32, 24, 16, 8]	0.8747	0.9176	0.9158	0.5031	0.9320

Table 4
The performance comparison with different filter sizes.

Filters	Sensitivity	Specificity	Accuracy	MCC	AUC
128	0.8695	0.9242	0.9220	0.5253	0.9298
256	0.8739	0.9037	0.9026	0.4787	0.9275
512	0.8578	0.9322	0.9192	0.5322	0.9339
1024	0.8746	0.9153	0.9139	0.4955	0.9305
2048	0.8621	0.9234	0.9208	0.5097	0.9299

Table 5
The performance comparison with different pre-trained language model features.

Feature sets	Sensitivity	Specificity	ACC	MCC	AUC
Cross-validation (5-folds)					
ProtTrans	0.8578	0.9322	0.9192	0.5322	0.9339
ESM-2	0.8506	0.9328	0.9286	0.5287	0.9208
ESM-1b	0.8904	0.8614	0.8625	0.4073	0.9186
Tape	0.8756	0.8948	0.8940	0.4597	0.9154
Independent test					
ProtTrans	0.9016	0.8749	0.8760	0.4331	0.9323
ESM-2	0.8852	0.8935	0.8931	0.4571	0.9222
ESM-1b	0.8852	0.8495	0.8511	0.3879	0.9115
Tape	0.8196	0.8942	0.8910	0.4236	0.8919

outperforms traditional classifiers such as KNN, RF, and SVM that use ProtTrans profiles. While the best-performing conventional method, RF, achieved an MCC of 0.5943 and an AUC of 0.8829, its low sensitivity of 0.4098 indicates that the RF method is more likely to miss true neoantigen epitopes during detection. In contrast, the multi-window CNN model, although achieving a lower MCC of 0.4331, has higher sensitivity (0.9016) and AUC (0.9323). This indicates that the multi-window CNN model is more effective at detecting true neoantigen epitopes and distinguishing between neoantigen epitopes and normal antigen epitopes. With this notable improvement in performance, ProtTrans embeddings are demonstrated to be highly effective when combined with multi-window CNN architectures. This approach allows better exploration of local and global patterns within sequence data, which leads to better predictive results in complex bioinformatics tasks. Fig. 4b shows the ROC curve of multi-window CNN compared to other classifiers.

3.6. Predictive analysis of peptides presented by MHC class I and II

We have separated the peptides presented by MHC-I and MHC-II from the independent dataset, as shown in Table 1, creating two distinct datasets, which include 18 MHC-I presented neoantigen epitopes and 43 MHC-II presented neoantigen epitopes separately. These datasets were then used in our model to perform a more focused predictive analysis, allowing us to evaluate the model's performance on each MHC class separately and better understand their unique peptide presentation pathways.

As shown in Table 7, it is worth noting that MHC-II has outperformed MHC-I in several metrics. Notably, MHC-II achieved a higher MCC value (0.4775) compared to MHC-I (0.3673), indicating a more balanced performance. Additionally, MHC-II demonstrated a higher AUC value (0.9055) than MHC-I (0.8893), reflecting robust model performance. This improved performance could be attributed to the longer sequence lengths of peptides typically presented by MHC-II compared to MHC-I. The longer sequences may allow the pre-trained language model used in our approach to extract more biological information, enhancing the model's ability to distinguish between different peptides presented by MHC-II.

3.7. Comparative analysis: prediction of MHC class II neoantigen epitopes

The prediction of MHC-II molecules is often considered to be a challenging task, primarily due to the diversity of MHC-II molecules and their complex binding mechanisms. Based on our independent test dataset, we isolated the MHC-II data and conducted a focused screening, resulting in 43 neoantigens epitopes that could be compared, demonstrating the superior performance of our model in MHC-II presented neoantigen epitopes prediction. The following are the state-of-the-art prediction methods included in our benchmark:

NetMHCIIpan [9]: This model is used to predict the binding affinity of all known HLA class II molecules. This model is trained on a large immunogenic dataset and is capable of predicting the binding affinity of different HLA class II molecules to antigens.

MixMHC2pred [10]: MixMHC2pred is a more specialized tool that aims to predict the diversity of MHC II molecules. By combining machine learning and immunoproteomics techniques, the model accurately predicts the binding characteristics of different MHC II variants and their specific ligands.

As shown in Table 8, we selected six neoantigen epitopes from the new MHC-II neoantigen epitopes dataset, along with their prediction scores from these two different models. According to the result shown in the table, our method missed only one out of the six neoantigens, whereas the two benchmark models, NetMHCIIpan and MixMHC2pred, correctly predicted only one.

Although the other two models predicted these epitopes would not bind to MHC-II molecules, these epitopes remain experimentally validated neoantigens capable of eliciting an immune response. As a result, the results presented by our model are more relevant. Thus, whereas other models failed to predict the immunogenic potential of these epitopes, our approach was able to capture these potential immunogenic bindings, enabling more precise identification of neoantigens. In tumor immunotherapy, neoantigens play a crucial role in activating T cells. Even if certain epitopes do not bind to MHC-II, that does not mean that they cannot be used as therapeutic targets. Our model's predictions emphasize the immunogenic potential of these epitopes, making them valuable for cancer vaccine design or immunotherapy applications.

3.8. Extend research: identifying high-potential targets for MHC class II presentation in cancer immunotherapy

Furthermore, using our DeepNeoAG model, we identified 22 potential neoantigen from the VDJDB [32] self-antigen dataset, specifically focusing on peptides presented by MHC class II molecules. As shown in Table 9, we input these 22 peptides into NetMHCIIpan [9], a peptide-MHC II binding prediction tool, to perform a score comparison. Our method predicted neoantigen peptides with high immunogenic potential, while NetMHCIIpan evaluated peptide-MHC II binding, with scores <5 indicating strong binding.

Upon comparison, some peptides (i.e., GELIGILNAAKVPAD, PGVLLKEFTVSGNIL, ENPVVHFFKNIVTP, among others) demonstrated both high neoantigen potential and strong binding affinity. This suggests that these peptides have the highest likelihood of being effective targets for cancer immunotherapy, as they exhibit both strong immunogenic properties and a strong ability to bind to MHC class II molecules. Their dual characteristics make them prime candidates for further experimental validation and potential inclusion in vaccine development or adoptive T-cell therapies.

However, some peptides exhibited high neoantigen potential but weaker binding according to NetMHCIIpan. For example, 'MTEYKLVVVGAVGVGKSALTIQLI' was labeled as neoantigen in our method, but received a relatively weaker binding score of 13.74 by NetMHCIIpan. These results highlight the complementary strengths of the two methods, where peptides with high immunogenic potential may not always exhibit the strongest binding. We believe these results may help guide future research in neoantigen discovery and selection, as they

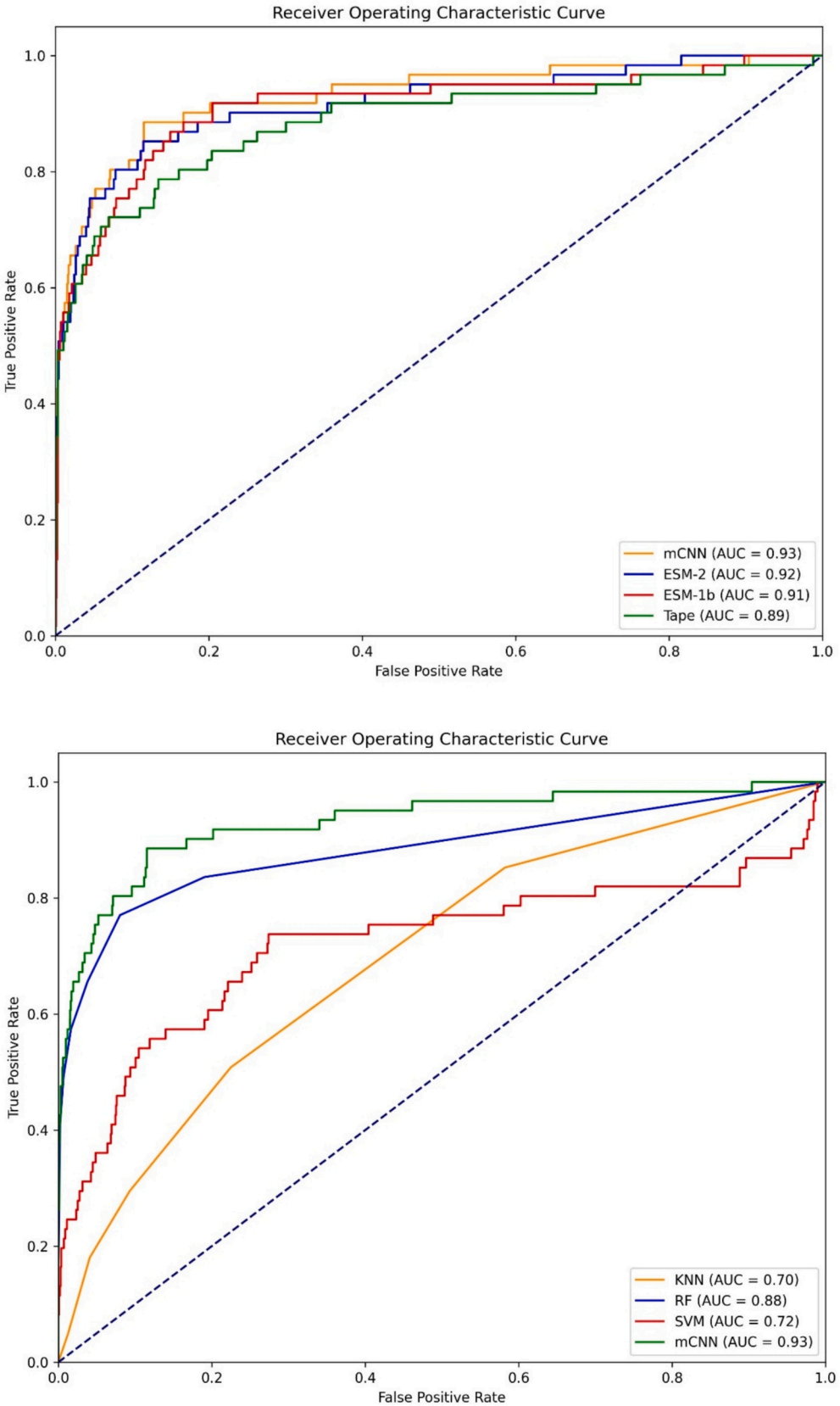


Fig. 4. (a) ROC Curve for the performance of predicting neoantigen epitope with with different pre-trained language model features. (b) ROC Curve for the performance of predicting neoantigen epitope with different classifiers.

Table 6
The performance comparison with different classification methods.

Classifier	Sensitivity	Specificity	Accuracy	MCC	AUC
KNN	0.2950	0.9069	0.8803	0.1361	0.6989
RF	0.4098	0.9977	0.9722	0.5943	0.8829
SVM	0.7377	0.7259	0.7265	0.2074	0.7203
mCNN	0.9016	0.8749	0.8760	0.4331	0.9323

Table 7
The performance of neoantigen prediction presented by MHC class I and II.

Dataset	Sensitivity	Specificity	Accuracy	MCC	AUC
Independent test	0.9016	0.8749	0.8760	0.4331	0.9323
MHC-I	0.7778	0.9413	0.9384	0.3673	0.8893
MHC-II	0.8605	0.8077	0.8136	0.4775	0.9055

Table 8
The performance comparison with previous study using MHC-II presented peptides from the independent testing data.

MHC-II presented peptides	Our methods	NetMHCIIpan	MixMHC2pred
	Neoantigen/non-neoantigen	MHC-II binding (<5 %)	
NNSKKKWFLFQDSKK	Neoantigen (O)	7.840 (X)	36.6 (X)
DFSQEFSPIQEAQQD	Neoantigen (O)	5.020 (X)	11.5 (X)
FLLTDYALSPDGSIR	Neoantigen (O)	34.18 (X)	60.3 (X)
FEAEFTQVAKKEPDG	Neoantigen (O)	58.96 (X)	53.7 (X)
DRASFLLTDYALSPD	Neoantigen (O)	0.420 (O)	1.89 (O)
VIFKSIMRQRLISPQL	Non-neoantigen (X)	6.6 (X)	11 (X)

Table 9
The performance comparison with previous study using MHC-II presented peptides from the VDJDB.

MHC-II presented neoantigen peptides for VDJDB	Our method	NetMHCIIpan
	Neoantigen/non-neoantigen	MHC-II binding (<5 %)
MTEYKLVVVGAVGVGKSALTIQLI	Neoantigen (O)	13.74 (X)
GELIGILNAAKVPAD	Neoantigen (O)	0.03 (O)
GELIGTLNAAKVPAD	Neoantigen (O)	0.14 (O)
KALARALKEGRIR	Non-neoantigen (X)	5.58 (X)
PGVLLKEFTVSGNIL	Neoantigen (O)	4.15 (O)
RNTFRHSVVVPCE	Neoantigen (O)	3.87 (O)
QPLALEGSLQKRG	Non-neoantigen (X)	19.25 (X)
LQPLALEGSLQKRG	Non-neoantigen (X)	16.46 (X)
FSWGAEGQRPGF	Non-neoantigen (X)	53.22 (X)
FSWGAEGQRPGFGSGG	Neoantigen (O)	61.54 (X)
LLEFYLAMPFATP	Neoantigen (O)	29.71 (X)
SLFFSAQPFEITAST	Neoantigen (O)	3.53 (O)
DPPALASTNAEVT	Non-neoantigen (X)	5.35 (X)
GYNSYSVSNSEKHIM	Neoantigen (O)	5.59 (X)
NHVVDISKSGLITIA	Neoantigen (O)	1.14 (O)
ATGFKQSSKALQRPVAS	Non-neoantigen (X)	7.39 (X)
GAVGVGKSAL	Non-neoantigen (X)	100 (X)
VVRHCPHHERCSDSD	Neoantigen (O)	43.43 (X)
ENPVVHFFKNIVTP	Neoantigen (O)	0.08 (O)
HYNYMCNSSCMGSMN	Neoantigen (O)	5.53 (X)
VVVGAVGVGKSALTIQLIQN	Neoantigen (O)	61.34 (X)
VHFFKNIVTPRTPG	Neoantigen (O)	2.1 (O)

emphasize the need to consider both immunogenic potential and binding affinity when identifying promising candidates for cancer immunotherapy.

4. Conclusion

In conclusion, our research represents DeepNeoAG, a deep learning model that combines protein language models and multi-window convolutional neural networks (MCNNs) to interpret tumor neoantigen sequences. One key advantage of DeepNeoAG is its ability to predict neoantigens across a broad range of MHC alleles without relying on specific allele information, thus reducing the need for multiple models and lowering complexity and cost. By directly capturing neoantigen sequence patterns, DeepNeoAG facilitates the identification of immunogenic neoantigens. Expanding its dataset to encompass a broader range of cancer types and integrating MHC binding prediction could further improve the model's accuracy and extend its applicability, potentially accelerating the development of neoantigen vaccines and personalized immunotherapy.

Through the use of pre-trained language models such as ProtTrans (ProtT5), ESM-2, ES-1b, and Tape, our study effectively analyzes antigenic epitopes and allows the model to consider contextual information within protein sequences and determining the relationships and significance of various amino acids autonomously. With this approach, we demonstrate the potential for advanced language models to provide precise predictions of protein functions and interactions in bioinformatics. By conducting a detailed comparison with traditional machine learning classifiers and utilizing various pre-trained language models, the integration of multi-window convolutional neural networks (mCNNs) with ProtTrans has demonstrated its effectiveness as a bioinformatics tool. This approach facilitates the precise identification of neoantigens, showcasing its potential in advancing cancer immunotherapy research.

Our findings emphasize the significance of using this advanced model in personalized cancer immunotherapy, offering the potential to develop more accessible and cost-effective treatments based on neoantigens. Beyond melanoma research, the classification of neoantigens has broad applications, ranging from the enhanced understanding of tumor biology to the exciting potential of predicting immunogenic shared neoantigens across various cancer types. This new approach to cancer treatment could revolutionize the field by enabling more precise targeting of cancer cells while minimizing adverse effects on normal tissues. Ultimately, this could lead to significantly improved outcomes for cancer patients, transforming the landscape of oncology and providing a roadmap for future therapeutic developments.

However, there are still some potential limitations to our model. One major challenge is the insufficiency of training data, as the performance of DeepNeoAG heavily depends on the quantity and quality of the data. The current dataset of tumor neoantigens is relatively small, which restricts the model's prediction accuracy and generalization ability. Additionally, the interpretability of the model needs improvement, as it remains unclear which sequence features of tumor neoantigens the protein language models (PLMs) have learned and how these features influence the prediction results. Improving these limitations will be a key issue in our future research on predicting neoantigens across diverse cancer types and MHC alleles.

CRediT authorship contribution statement

Cheng-Che Chuang: Writing – original draft, Software, Methodology, Conceptualization. **Yu-Yen Ou:** Visualization, Validation, Formal analysis, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Additional files

The code and data for this work are available at https://github.com/B1607/DeepNeoAG.

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